

Hello Mark Lights

Woodstock, GA

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<https://markjarredlights.com/home>

<https://github.com/DevMarkLights>

<https://marks-pi.com/>

Summary

Full-stack Software Engineer with 3+ years of experience designing and deploying scalable applications using **Java, Python, React, and SQL**. Currently pursuing an **M.S. in Computer Science (Machine Learning concentration)** at **Georgia Tech**. Skilled in **Spring Boot, REST API development, and cloud integration (Azure, GCP)**. Passionate about bridging software engineering and machine learning to build intelligent, production-ready systems that drive business impact..

Skills

Java, Javascript, python, HTML, Rest Apis, CSS, Spring Boot, React.js, Express.js, Node.js, Next.js, SQL, MongoDB, Azure, and Google Cloud, Git, Pandas, Scokit-Learn, PyTorch, Machine Learning

Experience

JUNE 2022 - PRESENT

General Motors, Roswell, GA - *Software Engineer 6*

Responsibilities:

- **Led production deployments across 4 enterprise applications**, coordinating with QA, DevOps, and business stakeholders to deliver releases with minimal downtime and consistent execution in high-availability environments
- Documented production deployment and rollback procedures to support release coordination and reduce recovery time during production issues
- Delivered full-stack enhancements using **Spring Boot** and **JSP**, improving user-facing workflows and supporting scalable feature delivery across business-critical systems
- **Designed and implemented batch interfaces leveraging Spring Batch**, optimizing data processing workflows and improving throughput for high-volume jobs with concurrency.
- Built and tuned complex **SQL queries, tables, stored procedures, and triggers**, improving query performance and ensuring data integrity across transaction-heavy workflows.
- Managed the production incident queue, resolving **P1/P2 outages**, performing root cause analysis, and implementing long-term fixes to reduce repeat incidents and improve SLA compliance.
- Developed and maintained **REST APIs** in Spring Boot, integrating internal microservices and downstream systems with **OAuth2 authentication** to support secure, business-critical workflows.

Education

JAN 2024 - May 2026

Georgia Institute of Technology, Atlanta, GA – *M.S. in Computer Science*

Concentration: Machine Learning

Classes:

- CS6035 : Introduction to Information Security
- CS7641 : Machine Learning
- CS6601 : Artificial Intelligence
- CS6603 : AI Code and Ethics
- CS6300 : Software Development Process
- CS7646 : Machine Learning for Trading
 - Built a reinforcement learning model to analyze stock price patterns and develop optimal trading strategies. This project deepened my understanding of AI-driven decision-making in financial markets
- CS6310 : Software Architecture and Design
 - Designed and built a Pokemon battle application where two pokemon can fight each other in an arena. Modeled with UML and built with Java.
- CS7650 : Natural Language Processing (LLM)
 - Built a large language model using pytorch. Used recurrent Neural Networks with LSTM cells.
 - Text parsing for building document vocabularies and prompt engineering
- CS7643 : Deep Learning
 - Built a MLP with softmax function from scratch using python and numpy
 - Learnt Transformer architecture and Convolutional Networks

AUG 2015 - May 2022

Francis Marion University, Atlanta, GA – *B.S. in Computer Science*

Projects

FEB 2026 - FEB 2026

Compress PDF Free

For anybody who gets tired of searching for a pdf compressor and you upload your files. Just to find out it requires your email to download it. Or worse you have a set limit on how many you can upload and it charges you a fee. This is for you. It's simple, free, and unlimited. This is hosted on my raspberry pi 5 at (<https://marks-pi.com/compressPDFApp/>) and all files uploaded are instantly deleted after being compressed and returned back to you. There is a catch only 1 file at a time.

Dec 2025 - Present

RAG Pipeline

Developed a document retrieval and question-answering application that allows users to upload PDF and TXT files and query their contents using semantic search. Implemented a Python/FastAPI backend utilizing the all-MiniLM-L6-v2 embedding model and ChromaDB for vector storage and similarity search. Built a React/Express frontend for user interaction and deployed the system on a self-hosted Raspberry Pi 5 environment (<https://marks-pi.com/rag>).

May 2025 - June 2025

AI-assistant

Developed a voice-enabled AI assistant powered by a locally deployed LLaMA 3.2 Instruct model. Implemented real-time speech-to-text and text-to-speech to enable hands-free, conversational interaction via microphone and speakers. Designed the system for low-latency, on-device inference, creating an immersive and privacy-focused AI experience.

GitHub: <https://github.com/DevMarkLights/Ai-Assistant>

NOV 2023 - NOV 2025

Lights Finance

Developed a finance web application that analyzes portfolio performance and income generation. Features include sector diversification analysis, monthly dividend calculations, and future portfolio value projections based on historical metrics. Integrated options rate-of-return analysis, unusual options activity insights, and a Robinhood report parser to compute monthly income from dividends and options. Self-hosted on a Raspberry Pi environment (<https://marks-pi.com/finance>).

AUG 2025 - OCT 2025

YahooFinance API

This is a rest api running locally on my raspberry pi 5. Lights Finance calls this api to get all stock and option information. It is built using the flask framework.